

SlabMaster User Guide & Reference

Welcome to the official documentation for **SlabMaster**, the national multi-tenant enterprise solution for countertop fabrication, slab inventory management, job scheduling, and mobile field dispatch.

Architectural Foundations

SlabMaster is designed from the ground up to solve the operational hurdles faced by high-volume commercial stone fabricators:

- **3-Tier Builder Hierarchy:** Direct modeling of `Account → Community → Lot` eliminates flat job naming hacks.
- **Multi-Facility Plant Scoping:** Independent facility hubs (e.g. `ATL`, `PHX`, `TUC`) with dedicated lead times, holiday calendars, and machine queues.
- **Offline PWA Engine:** Field crews can execute inspection checklists, capture customer signatures, and snap installation photos even with zero cellular connectivity.



Accounts & Communities

The Accounts module manages master relationships with commercial production homebuilders, general contractors, and designers.

The 3-Tier Enterprise Model

1. **Builder Account:** High-level corporate client (e.g., Lennar, Pulte, D.R. Horton) storing corporate billing terms and regional contacts.
2. **Subdivision / Community:** Master subdivision under the builder (e.g., *Silver Creek Ranch*) containing location coordinates, superintendent contacts, and default pricing sheets.
3. **Lot / Unit:** Individual residential or commercial unit with specific plan elevations and architectural configurations.

Jobs & Field Dispatch

Every job order in SlabMaster represents an actionable countertop installation milestone, automatically grouped by subdivision and lot.

Contractor Data Shielding

External subcontractors and field crews only see the job site address, lot number, and technical specs necessary to complete the install. Commercial margins, PO amounts, and billing records are strictly shielded.

Activity Sequencing

Jobs track four sequential milestones:
**Laser Template → CAD / CAM
Programming → Shop Fabrication →
Field Installation.**

Production Calendar & Auto-Scheduling

SlabMaster features an automated 14-day scheduling engine that sequences fabrication and installation milestones while strictly enforcing working days and regional facility shutdowns.

- **Working-Day Rules:** Scheduling offsets skip Saturdays, Sundays, and official company holidays.
- **Plant Holiday Overrides:** Configure regional shutdowns per operating facility (e.g., weather emergencies or machine maintenance).
- **Touch Swipe & Stacked Views:** Touch gestures on mobile devices allow shop managers to swipe day-by-day or inspect hourly crew distributions.

Slab Inventory & Remnant Management

Full tracking of raw stone materials, bundle allocations, warehouse rack coordinates, and remnant offcuts.

Key Features

- **Dimensional SQFT Calculation:** Enter Length (in) and Width (in) to calculate total square footage instantly: $SQFT = (L \times W) / 144$.
- **Remnant Offcut Linkage:** Tag remnants with parent slab serial numbers to track yield utilization and residual cost basis.
- **A-Frame Bin Tagging:** Assign slabs to specific physical warehouse racks (e.g., Rack B-04).
- **Thermal Barcode Printing:** Generate scannable QR and Code-128 barcode labels for thermal shop printers.

Purchasing & Dock Receiving

Streamline vendor purchase orders for raw slabs, sinks, and consumable tooling with integrated dock receiving.

One-Click Dock Receiving

When a slab bundle arrives at the receiving bay, shop managers can review the PO line items and click **Dock Receive**. Slabs are automatically assigned unique serial numbers, allocated to storage racks, and marked **Available** in inventory immediately.



Shop Floor Kiosk View

A high-contrast, touch-optimized station monitor designed for fabrication machine operators (Sawjet, CNC, and Polish stations).

Machine Station Queues

- **Saw 1 (Bridge Saw):** Initial straight-line perimeter and sink cutout rough cuts.
- **Saw 2 (Waterjet CNC):** High-precision curve cuts, miters, and complex island cuts.
- **CNC Router:** Edge profiling, polishing, and undermount sink drop shaping.
- **Polish Line:** Hand finishing, backsplash chamfering, and final QC inspection.

Operators can advance a job from *Cut Queue* → *In Progress* → *Complete* with a single tap on ruggedized shop tablets.



Dynamic Custom Attributes Engine

Define multi-tenant custom schema attributes at runtime without code deployment.

Entity Targeting & Data Types

Custom attributes can be attached to `JOB` , `ACCOUNT` , `ACTIVITY` , or `SLAB` entities.

Data Type	Description	Example Use Case
<code>text</code>	Freeform text string	Project Coordinator Name
<code>currency</code>	Monetary USD values	Invoice Amount / ERP Sync
<code>select</code>	Configurable dropdown choices	Fabrication Change Reason
<code>boolean</code>	Toggle switch	Crane Lift Required
<code>date</code>	Calendar date selector	Builder Frame Inspection Date
<code>url</code>	Clickable external hyperlink	2020 Design / CAD Cloud Link



Table Matrix Sub-Grids & SAP Config Sheets

Moraware configuration parity allowing embedded multi-row, multi-column sub-tables inside custom inspection and sign-off forms.

SAP Configuration Sheet (Template `ft_sap_config_sheet`)

Enables fabricators to enter room-by-room stone specifications (Kitchen, Master Bath, Island) with automated column summation for Cutout Counts and Splash Lineal Feet.



Offline PWA Form Packets

Build custom form templates, bundle them into multi-stage packets, and execute them on field tablets with offline IndexedDB storage and touch signature capture.

Multi-Plant & Security Settings

Configure regional operating facilities, lead time matrices, holiday closures, Entra ID SSO policies, and white-label subscriber branding.

Modern RESTful API & SAP ERP Integration

SlabMaster provides a modern JSON RESTful API designed to supersede legacy SOAP and .NET interfaces with full CRUD control and first-class External ID routing.

Key API Capabilities

- **External ID Routing:** Directly query, upsert, and link Builder Accounts, Communities, Lots, Jobs, and Job Activities using your SAP ERP identifiers (WBS Elements, PO numbers) without translating internal UUIDs.
- **Two-Way Change Data Capture (CDC):** Poll `GET /api/v1/sync/changes?since=<timestamp>` to mirror shop floor completion, field measure signoffs, and install progress back to SAP in real time.
- **Audit-Compliant Soft Cancellation:** `DELETE` endpoints idempotently set `status: "Cancelled"` and `isArchived: true` to avoid breaking downstream foreign keys in SAP cost collector accounting.
- **Admin API Key Management:** Generate scoped tokens (`sm_live_...`) directly in the Admin Portal under Settings → API & ERP Integration.



Download Postman Collection (.json)



Subscriber Onboarding Checklist & Initial Setup Guide

Comprehensive blueprint for provisioning and configuring a new fabricator or enterprise subscriber tenant from zero to live production.

Part 1: New Subscriber Information Checklist

Gather the following prerequisite operational and system parameters prior to tenant provisioning:

- **Organization & Identity:** Legal company name, tenant domain (e.g. `granitecraft.com`), high-res transparent logo, and authentication mode (Entra ID SSO or Hybrid/Local).
- **Regional Facilities & Plants:** Facility names, plant codes (e.g. `ATL`, `PHX`), physical yard addresses, timezones, and holiday closure calendars.
- **Crews & Machinery (Assignees):** CNC Sawjet lines, CNC routers, digital laser templating techs, installation teams, and warranty repair trucks.
- **Activity Lifecycle Pipeline:** Standard phase progression (Template → CAD Match → Sawjet → Polish → Field Install → Sign-off) with durations and lead time dependencies.
- **Material & Slab Catalog:** Stone types (Quartz, Granite, Porcelain), color brands (Cambria, Silestone, MSI), standard slab dimensions, and minimum remnant thresholds.
- **Builder Accounts & Job Sites:** Top-level builder accounts, community subdivisions, lot numbers, and corresponding SAP/ERP External IDs (e.g. `SAP-CUST-126954`, `SAP-COMM-LNXAUS`, `SAP-LOT-LNXAUS-036`).
- **Form Templates & Sub-Grids:** Inspection checklists, room-by-room Table Matrix sheets (e.g. `ft_sap_config_sheet`), and touch glass sign-off waivers.
- **ERP Integration Credentials:** SAP S/4HANA or Moraware endpoints, REST API key scopes, and outbound retry destination URLs.

Part 2: 10-Step Chronological Setup Sequence

1. **Tenant Provisioning:** Create subscriber tenant profile, upload company logo, and configure Entra ID / Local administrator accounts.
2. **Regional Hub Configuration:** Register operating facilities (e.g. Atlanta Metro Hub), set local timezones, and define operating hours / company holidays.
3. **Register Stations & Mobile Crews:** Add CNC sawjet machines and field mobile trucks as assignees with unique schedule color codes.
4. **Build Activity Pipeline:** Create standard milestones (Laser Template, CAD Match, Sawjet Cut, Install, QA Signoff) with inter-activity lead time buffers.
5. **Load Slab Inventory:** Enter initial stone materials, colors, bundle numbers, and barcode IDs (e.g. `SLB-ATL-2026-0041`) with bin/rack warehouse locations.
6. **Setup Builder Master Hierarchy:** Create the first Builder Account (e.g. Lennar GA), Subdivision Community (Austin Park), and Lot (Lot 000036) with ERP External IDs.
7. **Configure Table Matrix & Forms:** Build digital form templates, room specification sub-tables, and touch signature fields for field tablets.
8. **Create First Job & Schedule:** Create the first work order (e.g. `SAP-S0-10170`), allocate warehouse slabs, and apply the automated activity schedule template.
9. **Test Shop Floor & Mobile Sign-Off:** Simulate shop floor kiosk cut execution on Sawjet Line 1 and mobile PWA offline signature capture.
10. **Generate API Keys & Connect ERP:** Generate live REST API credentials (`sm_live_...`) and verify two-way sync with the Outbound ERP Retry Queue.



Video Walkthroughs & YouTube Tutorial Channel Pack

High-definition (1080p) screen recordings featuring glowing cursor tracking, click ripple indicators, and dynamic chapter banners. All videos are stored in the project's `videos/` directory and paired with full voiceover cue sheets.

Master Subscriber Setup Walkthrough (1080p)

Demonstrates the complete end-to-end subscriber onboarding flow from Day Zero to live shop floor production and enterprise ERP integration.

- **Video File:**

`videos/01_Master_Onboarding_Walkthrough/full_subscriber_setup_walkthrough.webm`

- **Narration Script & Timestamps:**

`videos/01_Master_Onboarding_Walkthrough/full_subscriber_setup_walkthrough_transcript.md`

- **Chapters:** Tenant Provisioning → Regional Hubs (ATL) → Machinery (Sawjet Line 1) → Barcode Inventory → 3-Tier Builder Hierarchy (Lennar → Austin Park → Lot 000036) → Work Order SAP-SO-10170 → Production Calendar → Kiosk Touch Queue → Offline PWA Forms → REST API Tokens & Dead-Letter Recovery Queue.

Feature Deep-Dive Video Library

Dedicated high-contrast recordings and YouTube shorts for each modular feature:

- **01_builder_hierarchy_accounts_communities.webm:** 3-Tier builder hierarchy, community subdivisions, lots, and SAP customer IDs.
- **02_jobs_orders_and_scheduling.webm:** 6-stage lifecycle milestones, lead times, and multi-plant working-day engine.
- **03_slab_inventory_and_remnants.webm:** Serialized barcodes, rack bin locations, and remnant threshold management.
- **04_purchasing_and_dock_receiving.webm:** Distributor purchase orders, ETA tracking, and one-click dock receiving.
- **05_shop_floor_kiosk_stations.webm:** High-contrast touch tablet queue for CNC sawjets and active cut timers.
- **06_table_matrix_and_offline_forms.webm:** Field measure sheets, dynamic room sub-grids, and touch glass stylus signatures.
- **07_admin_settings_api_and_erp_queue.webm:** Scoped REST API keys, progressive exponential backoff ladder, and dead-letter monitoring.
- **Voiceover Scripts:** `videos/02_Feature_Deep_Dives/feature_videos_voiceover_scripts.md`